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ZECPATH BACKEND SYSTEM DEBUGGING REPORT

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Date / Timestamp : 2026-09-30T18:40:00Z
Auditor : Backend Engineering Team
Scope : Production Stabilization, ORM Tuning, Defensive Fault-Tolerance
Deliverable Path : docs/day_64/DEBUGGING_REPORT.md

1. EXECUTIVE SUMMARY

- Total Defects Evaluated : 6
- Critical Severity (P1) : 2 (Deadlocks, OOM Worker Termination)
- High Severity (P2) : 2 (N+1 Performance Degradation, Route 404s)
- Medium Severity (P3) : 2 (Stateless JWT Role Drift, NullType Arithmetic)
- Resolution Rate : 100% Resolved and Verified

2. VERIFICATION BENCHMARKS

- Endpoint: /api/recruiter/premium/jobs/1/ranking-report/
 - * Pre-fix Latency : 2,840 ms (201 DB Queries)
 - * Post-fix Latency : 18 ms (2 Consolidated Queries via select_related)
- Celery Worker Memory:
 - * Pre-fix Behavior : Continuous memory growth (OOM crash after ~40 resumes)
 - * Post-fix Behavior: Stable at <120 MB RSS with max-tasks recycling active
- Route Resolution:
 - * Corrected path mapping verified with HTTP 200 OK responses in Postman

3. SYSTEM SIGN-OFF

- Defensive type casting implemented across all scoring formulas.
- Pessimistic locking active on all subscription state transitions.
- All tests passing with no functional or performance regressions.